

DETAILED DESCRIPTION OF THE INVENTION — CALIBRATION, CONTACT CONFIRMATION, AND DATA ACQUISITION METHODS

In various embodiments, the intraluminal physiologic monitoring device may employ calibration procedures configured to establish individualized measurement baselines and ensure reliable physiologic signal acquisition. Calibration routines may be executed automatically upon insertion, upon user initiation, periodically during operation, or according to predefined timing schedules.

Calibration may include establishing electrical reference baselines for impedance or capacitive sensing, thermal stabilization measurements, pressure normalization, orientation reference determination, and signal quality verification. Initial calibration may allow the device to adapt to individual anatomy, insertion depth, and environmental conditions present during measurement.

Contact confirmation mechanisms may be implemented to verify appropriate positioning and tissue interaction prior to physiologic data acquisition. Contact confirmation may be inferred through analysis of impedance stability, pressure signatures, dielectric response, motion behavior, temperature convergence, or combinations thereof. The system may determine when sufficient contact stability exists to initiate valid measurement sessions.

In certain embodiments, the device may continuously evaluate signal integrity and dynamically adjust sampling parameters based on detected motion, loss of contact, environmental disturbance, or physiologic variability. Adaptive sampling techniques may include modification of sampling frequency, sensor activation sequences, filter parameters, or wireless transmission intervals.

Data acquisition may occur continuously, intermittently, or event-triggered depending on device configuration and intended use. Acquisition modes may include session-based measurement, periodic monitoring, scheduled sampling windows, or adaptive measurement triggered by detected physiologic conditions.

Artifact rejection techniques may utilize motion sensing, signal redundancy, statistical filtering, or machine learning approaches to remove invalid measurements caused by user movement, improper placement, or transient environmental factors. Quality scoring metrics may be generated to characterize confidence levels associated with collected data.

Collected physiologic signals may undergo preprocessing within onboard electronics prior to transmission. Preprocessing operations may include amplification, filtering, digitization, normalization, feature extraction, compression, encryption, or anonymization.

Calibration and acquisition methods described herein enable repeatable longitudinal measurement across multiple insertion events while minimizing user burden and improving data consistency over time. Alternative calibration strategies, confirmation techniques, and acquisition architectures capable of achieving substantially similar functional outcomes are considered within the scope of the disclosure.